| Chapter | Title | Total questions | Type |
|---|---|---|---|
| Ch 8 | **Journey Inside the Atom** | 12 | Formative |

This assessment is formative. Every question traces to a specific learning outcome, and each outcome is derived from an NCF competency this chapter develops. Rather than testing recall or asking whether a student "understood" a topic, each item asks for an observable demonstration — a response, a classification, a justification, or a product — of the thinking the lessons built. The questions are grouped by sections, and each shows the learning outcome it assesses and its cognitive demand, so you can see exactly what each question checks and why.

SECTION 1
## 8.1 Rediscovering the Roots of Atomic Theory

**Q1** · **Learning Outcome**
Students can distinguish between philosophical reasoning and evidence-based scientific argument by identifying the specific evidential criteria that one framework provides and the other lacks.

**Cognitive Demand**
Understanding

Acharya Kanada and the Greek philosopher Democritus both proposed that matter consists of tiny, indivisible particles. Dalton, in 1808, also proposed indivisible atoms. What does Dalton's proposal have that the ancient accounts lack?

**A.** Dalton gave atoms specific names derived from Sanskrit and Greek roots.

**B.** Dalton was the first to propose that matter is made of indivisible particles, while Kanada and Democritus did not.

**C.** Dalton's atoms are based on experimental evidence and specific claims about how atoms of different elements combine in fixed ratios.

**D.** Dalton's atoms are spherical in shape, whereas ancient philosophers described atoms as having different geometrical forms.

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SECTION 2
## 8.2 A Short Historical Journey Through Atomic Models

**Q2** · **Learning Outcome**
Students can evaluate a scientific model by identifying the evidence that supports it and the questions it leaves unresolved, using an analogy to expose the model's structural assumptions.

**Cognitive Demand**
Analysis

Thomson's cathode ray experiment provided two key pieces of evidence that led him to propose the plum pudding model of the atom.

**(a)** State the two pieces of evidence and the inference each one supports.

**(b)** The plum pudding model is often compared to a watermelon — red pulp as positive matter, seeds as electrons. Identify one aspect of atomic structure the watermelon analogy represents correctly and one structural question about the atom the analogy cannot answer.

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SECTION 3
## 8.2.2 Testing Thomson's model: The gold foil experiment

**Q3** · **Learning Outcome**

**Cognitive Demand**

| | |
|---|---|
| Students can construct a logical argument showing how a single anomalous experimental result falsifies a model, and derive structural inferences about a system from the pattern of observed deflections. | Analysis |

In the gold foil experiment, most alpha particles passed straight through the foil, a small number deflected at large angles, and a very few bounced directly back. Which observation is sufficient on its own to falsify Thomson's plum pudding model, and what structural feature of the atom does it imply?

**A.** A few particles bouncing directly back — this implies a tiny, dense, positively charged nucleus, which Thomson's diffuse positive sphere cannot produce.

**B.** Most particles passing straight through — this shows the atom is mostly empty space, which Thomson's model did not predict.

**C.** Some particles deflecting at large angles — this shows electrons are negatively charged, confirming Thomson's placement of negative charges inside the atom.

**D.** The combination of all three observations together is required; no single observation alone is sufficient to falsify Thomson's model.

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| **Q4** · **Learning Outcome** | **Cognitive Demand** |
|---|---|
| Students can construct a logical argument showing how a single anomalous experimental result falsifies a model, and derive structural inferences about a system from the pattern of observed deflections. | Application |

Rutherford's planetary model drew three conclusions from the gold foil experiment. However, a physics principle requires a charged particle moving in a circular orbit to continuously radiate energy.

**(a)** State the three conclusions Rutherford drew.

**(b)** Use the physics principle stated above to explain why an electron orbiting the nucleus according to Rutherford's model would be unstable.

**(c)** A helium atom has 2 protons. State the number of electrons in a neutral helium atom and explain why.

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SECTION 4

## 8.2.3 Bohr's model of the atom

| **Q5** · **Learning Outcome** | **Cognitive Demand** |
|---|---|
| Students can explain how introducing the concept of fixed energy levels resolves a contradiction in a prior model by identifying the specific physical consequence the new postulate prevents. | Understanding |

Bohr proposed that electrons occupy fixed circular paths called stationary states, each with a definite constant energy. Which statement correctly explains how this postulate resolves the problem with Rutherford's model?

**A.** Bohr's shells are labelled K, L, M, N because earlier letters were reserved for a possible inner series, which prevents energy loss from the atom.

**B.** Electrons in Bohr's model can only move between shells when they absorb energy, so they naturally remain in the lowest shell and never spiral inward.

**C.** In a stationary state, the electron moves but does not radiate energy, so it maintains its orbit indefinitely without spiralling into the nucleus.

**D.** In a stationary state, the electron stops moving, so it no longer accelerates and does not radiate energy.

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SECTION 5

## 8.3 What Components Contribute to the Mass of an Atom?

| **Q6** · **Learning Outcome** | **Cognitive Demand** |
|---|---|
| Students can resolve a quantitative discrepancy in a scientific model by identifying the missing component and explaining the physical role that component plays in the system's stability. | Application |

Hydrogen has 1 proton and helium has 2 protons, yet helium is approximately four times as heavy as hydrogen, not twice as heavy.

**(a)** Identify the particle discovered by Chadwick that accounts for this discrepancy and state its charge and approximate mass.

**(b)** Uranium has 92 protons and 146 neutrons. Explain, using the information about neutrons from this chapter, why uranium needs far more neutrons than protons to remain stable.

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SECTION 6

## 8.4 Symbols of Elements

**Q7 · Learning Outcome**
Students can apply a rule-governed symbolic notation system to encode and decode quantitative information about atomic composition from a standardised representation.

**Cognitive Demand**
Application

The nuclide notation for an atom is written as $^{23}_{11}Na$. Using this notation, how many neutrons does this atom contain?

**A.** 11

**B.** 12

**C.** 23

**D.** 34

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SECTION 7

## 8.6 Mass Number

**Q8 · Learning Outcome**
Students can apply a multi-variable numerical relationship to determine unknown subatomic quantities, showing each step of the reasoning chain explicitly.

**Cognitive Demand**
Application

Refer to the table provided. Complete all missing values for each atom, showing the relationship you use for each row.

| Atom | Atomic Number (Z) | Mass Number (A) | Protons | Neutrons | Electrons |
|---|---|---|---|---|---|
| Carbon | 6 | 12 | ? | ? | ? |
| Sulfur | 16 | 32 | ? | ? | ? |
| Chlorine | 17 | ? | 17 | 18 | ? |
| Calcium | ? | 40 | 20 | ? | 20 |

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SECTION 8

## 8.7 How Are Electrons Distributed in Different Energy Levels?

**Q9 · Learning Outcome**
Students can construct the electron shell distribution of an atom by systematically applying capacity and filling-order rules, and identify the valence shell from the resulting configuration.

**Cognitive Demand**
Application

An atom has atomic number 16.

**(a)** Write the electronic configuration of this atom, showing the number of electrons in each shell.

**(b)** Identify the valence shell and state the number of valence electrons.

**(c)** A student writes the configuration of the atom with atomic number 18 as 2, 8, 8. Another student writes it as 2, 16. State which is correct and explain the rule that makes the second configuration impossible.

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SECTION 9

## 8.8 Combining Capacity of an Atom: Valency

**Q10 · Learning Outcome**

Students can predict the combining capacity of an atom by applying the octet rule to its electronic configuration and distinguishing between electron loss, gain, and sharing pathways.

**Cognitive Demand**

Application

Magnesium has atomic number 12. Its electronic configuration is 2, 8, 2. What is the valency of magnesium and what does the atom do to achieve a stable configuration?

**A.** Valency 2; magnesium gains 2 electrons, filling its M shell to 8.

**B.** Valency 2; magnesium loses 2 electrons, leaving a complete L shell as the new outermost shell.

**C.** Valency 6; magnesium gains 6 electrons to fill its outermost shell to 8.

**D.** Valency 8; magnesium shares 8 electrons to reach an octet.

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SECTION 10

## 8.9 A Deeper Look into Atomic Structure

**Q11 · Learning Outcome**

Students can classify atomic variants by their conserved and differing quantities, and calculate a statistically weighted average to represent a naturally occurring mixture of isotopic masses.

**Cognitive Demand**

Application

Refer to the table provided.

**(a)** Calculate the weighted average atomic mass of boron. Show your working.

**(b)** Does any single boron atom in nature have the mass you calculated? Explain your answer.

**(c)** Boron-10 and Boron-11 are isotopes of the same element. State two quantities that are the same for both atoms and one quantity that differs.

| Isotope | Mass Number | Natural Abundance (%) |
|---|---|---|
| Boron-10 | 10 | 19.9 |
| Boron-11 | 11 | 80.1 |

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**Q12 · Learning Outcome**

Students can classify atomic variants by their conserved and differing quantities, and calculate a statistically weighted average to represent a naturally occurring mixture of isotopic masses.

**Cognitive Demand**

Analysis

The chapter states that isotopes of an element have the same chemical properties but different physical properties. Argon (Z=18, A=40), potassium (Z=19, A=40), and calcium (Z=20, A=40) are described as isobars.

**(a)** Explain, using the concept of electronic configuration, why the two isotopes of carbon ($^{12}_{6}C$ and $^{13}_{6}C$) have identical chemical properties despite having different masses.

**(b)** Argon, potassium, and calcium are called isobars, not isotopes. Construct an argument — using specific values of Z and A — distinguishing why these three atoms cannot be isotopes of the same element, and what structural feature they share that makes them isobars.

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